

Amadeus, VISA

Q292 | LC 2187: Min time to complete trips

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Problem Understanding & Approach

- basically $\text{time}[i] \rightarrow$ time taken by bus i to complete 1 trip
- so we need to find min time such that all buses can complete t trips
- so if they can complete in t time, they can also do in $t+1, t+2...$
- so **Binary Search**
- $l = 1$ as time can't be 0 as $\text{time}[i] \geq 1$ (atleast 1)
- $h = ? \rightarrow$ suppose if the bus taking the shortest time reaches t trips then other buses don't need to do anything
- so $h = (\text{min_element}()) \times \text{total_trips}$
- now in check function, we simply see if every bus can reach t trips in given time

⚠ **EACH BUS DON'T NEED TO COMPLETE T TRIPS, THE TOTAL NUMBER OF TRIPS BY ALL BUS MUST BE T**

C++ Solution

```
class Solution {
public:

    bool check(long long sustime, vector<int>& time, int totalTrips) {

        long long currtrips = 0;
        for(int t : time) {

            currtrips += sustime / t;
            if(currtrips >= totalTrips) {
                return true;
            }
        }

        return false;
    }

    long long minimumTime(vector<int>& time, int totalTrips) {

        int n = time.size();
        long long l = 1;
        long long h = 1LL * (*min_element(time.begin(), time.end())) *
totalTrips;

        long long ans = h;

        while(l <= h) {
            long long mid = l + (h - l) / 2;
            if(check(mid, time, totalTrips)) {
                ans = mid;
                h = mid - 1;
            }

            else {
                l = mid + 1;
            }
        }

        return ans;
    }
};
```

